MIL-P-62342(AT)
24 December 1981

MILITARY SPECIFICATION

PANEL ASSEMBLY, INSTRUMENT: DRIVER'S, M1 TANK

This specification is approved for use by US Army Tank-Automotive Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an instrument panel assembly which displays engine and vehicle speeds, electrical system voltage, and fuel levels; and monitors status of 22 points of vehicle maintenance and fire response. Panel assembly contains fire extinguisher second shot control switch, and controls fuel transfer. Panel assembly is sealed against environment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the soliciation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-C-572	- Cord, Yarns and Monofilaments, Organic Synthetic Fiber.
MIL-C-22520	- Crimping Tools, Terminal, Hand or Power Actuated, Wire Termination, and Tool Kits, General.
MIL-E-45782	- Electrical Wiring, Procedures for.
MIL-R-46846	- Rubber, Synthetic, Heat Shrinkable.
MIL-M-81594	- Marking Foil, Hot Stamp Printing of Electrical Insulating Materials.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Tank-Automotive Command, ATTN: DRSTA-GSS, Warren, MI 48090, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document, or by letter.

FSC 2540

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STANDARDS

FEDERAL

FED-STD-H28/2	- Screw Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-202	- Test Method for Electronic and Electrical Component Parts.
MIL-STD-454	- Standard General Requirements for Electronic Equipment.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1184	- Electrical Components for Automotive Vehicles; Waterproofness Tests.
MIL-STD-1275	- Electrical Circuit, 28 Volts DC Transient Characteristics for Military Vehicles.
MS3367	- Strap, Tiedown, Electrical Components, Adjustable, Self-clinching, Plastic, Type I, Class 1.

2.1.2 Other Government documents, and Government drawings and publications. The following documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

11655194	- Process Specification for Soldering Electrical Connections for Electrical and Electronic Equipment, General Requirement for.
12282017	- Assembly; Driver's Instrument Panel.
12282019	- Schematic Diagram, Driver's Instrument Panel.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity, or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

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3. REQUIREMENTS

3.1 First article (preproduction). A first article sample of the instrument panel to be furnished under this specification shall be produced after award of contract and before beginning production. Sample shall be representative of units to be manufactured in quantity, and shall be constructed with tools and equipment normally used in quantity production. Sample shall be inspected by the contractor under surveillance of the Government in accordance with, and shall pass, first article inspection specified herein (see 4.4 and 6.3). After approval is granted, production items shall not deviate from sample without approval of the Government.

3.2 Materials. Materials shall be as specified herein and in referenced specifications, standards, and drawings. Material shall be free of defects which adversely affect performance or serviceability of the finished product (see 6.4).

3.3 Design and construction. Panel assembly shall conform to drawings 12282017, 12282019, and specifications, and standards as may be applicable. Components shall be interchangeable as to fit, form, and function with all other units bearing the same part number.

3.3.1 Standard parts. Military standard parts shall be incorporated whenever possible. Commercial standard parts may be used, provided they are interchangeable with military standard parts without modification.

3.3.2 Threaded parts. Screw threads of the form, number per inch, and class specified on the applicable drawing or military standard shall be in accordance with FED-STD-H28/2.

3.3.2.1 Locking devices. Lock washers, self-locking nuts, safety wires, or other approved locking devices shall be incorporated where specified, or where required, to prevent loosening of components.

3.3.3 Dissimilar metals. Dissimilar metals shall be protected from galvanic corrosion in accordance with requirements of MIL-STD-889.

3.3.4 Ozone resistance. When rubber components are required to be ozone resistant, the degree of resistance shall be as specified in applicable specifications or drawings.

3.3.5 Soldering. Where soldering is specified, electrical connections shall be soldered in accordance with drawing 11655194.

3.3.6 Crimped contacts. Crimp type connector contacts shall be crimped using tools and procedures defined in MIL-C-22520.

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3.3.7 Eye terminals. MS type eye terminals shall be crimped to conductor and insulation or conductor, where applicable, in accordance with MIL-E-45782.

3.3.8 Heat shrinking. Rubber tubing, conforming to MIL-R-46846, shall be uniformly heat shrunk until tubing configuration reflects contour of surface to be insulated.

3.3.9 Lacing. Wire bundles shall be bound using any of following options:

a. Spot tied, or laced with specified lacing tape, in accordance with method B of MIL-E-45782, except that every third stitch shall be double lock.
b. Secured with nylon cable straps conforming to MS3367. Straps shall be applied at intervals of 2 inches or less, and shall be installed so as not to damage wire insulation.
c. Woven nylon covering, conforming to MIL-C-572, type P, form Y, to conform to intent of method B of MIL-E-45782, except that covering shall not unravel or run, if cut or abraded, and shall not cause damage to wire insulation.

3.4 Performance.

3.4.1 Input voltage. Instrument panel assembly shall operate within 24 Volt direct current (Vdc) vehicle electrical system, and shall operate as specified herein during and after exposure to variations of voltage between 11 Vdc and 30 Vdc.

3.4.2 Voltage drop. Unless otherwise specified herein, voltage drop through master panel assembly shall be not more than 0.5 Vdc in any circuit.

3.4.3 Current capacity. Instrument panel assembly shall carry 250 $\pm$ 25 milliamperes (ma) at 28 Vdc.

3.4.4 Circuit resistance. Resistance in circuits, except those containing power utilizing elements, shall be not more than 1.0 ohm.

3.4.5 Circuit isolation. Individual circuits shall be isolated from each other and from panel housing so that resistance to crossover is not less than 100 megohms at 110 Vdc.

3.4.6 Transient voltage characteristics. Assembly shall conform to requirements of MIL-STD-1275 regarding ripple, surges, spikes, and starting disturbances in fault-free, single fault, and multi-fault conditions.

3.4.7 Lamps circuits. Each lamp shall light on application of correct power, of correct polarity, to appropriate panel terminals. Removing power

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shall extinguish lamp. Lamps shall glow at not less than 11 Vdc. Lamp brightness shall increase as voltage increases. Lamps shall withstand not less than 27 Vdc.

3.4.7.1 Independent circuits. Lamps listed below shall operate independently of engine starting circuits:

Engine oil low	Air cleaner filter
Cable disconnected	Fuel control faulty
Circuit breaker monitor	Fire
Engine oil filter	Fire extinguisher 1st shot discharged
Transmission oil filter	Low fuel level
Fuel water separator filter	Low battery charge

3.4.7.2 Dependent circuits. Lamps listed below shall operate only when engine STARTED circuit is closed:

Hydraulic system malfunction	Engine oil pressure low
Transmission oil low	Engine overspeed
Rear fuel pump inoperative, right	Gas overtemperature
Rear fuel pump inoperative, left	Transmission oil temperature high
Engine oil temperature high	Transmission oil pressure low

3.4.7.3 LOW BATTERY CHARGE lamp. LOW BATTERY CHARGE lamp shall light when input voltage is not more than 23.1 Vdc. Lamp shall light after not less than 100 and not more than 200 seconds elapse at low voltage input.

3.4.7.4 HYDRAULIC SYSTEM MALFUNCTION lamp. HYDRAULIC SYSTEM MALFUNCTION lamp shall light when function circuit opens and malfunction circuit closes. Lamps shall light when not less than 4 and not more than 6 seconds elapse after change of circuits.

3.4.8 Meters.

3.4.8.1 Tachometer. Tachometer shall indicate full scale within $\pm$ 5 percent when subjected to 5 Vdc $\pm$ 0.1 Vdc applied at appropriate panel terminals.

3.4.8.2 Speedometer. Speedometer shall indicate 60 kilometers per hour (km/h) $\pm$ 4.5 km/h when subjected to sine wave signal of 1812 Hertz (Hz) $\pm$ 15 Hz at 2V peak-to-peak applied at appropriate panel terminals.

3.4.8.3 Fuel meter. Fuel meter shall indicate full scale (Full) within $\pm$ 5 percent when subjected to 5 ma (+0, -0.1 ma) applied at appropriate panel terminals.

3.4.8.4 Voltmeter. Voltmeter shall indicate full scale (30 Vdc) within $\pm$ 5 percent when subjected to 30 Vdc $\pm$ 0.1 Vdc applied at appropriate panel terminals.

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3.4.9 Switches.

3.4.9.1 Fire extinguisher switch (second shot). When actuated in connection with 112 ohm ± 5 percent (10 watt) resistor, voltage drop through closed second shot switch shall be not more than ± 0.5 Vdc.

3.4.9.2 Fuel tank selector. When actuated, switch shall make indicated circuit, break all others between fuel tank transducer, LOW FUEL LEVEL warning lamp, fuel pump, and panel fuel meter, and pass current to indicate correct fuel level in each tank.

3.4.10 Climate. Instrument panel assembly shall operate as specified herein during and after exposure to extremes of temperature and humidity as follows:

3.4.10.1 High temperature. Instrument panel assembly shall withstand storage in ambient temperature of 160°F. Thereafter, assembly shall operate as specified herein during and after exposure at 140°F ambient temperature.

3.4.10.2 Low temperature. Instrument panel assembly shall withstand storage in ambient temperature of minus 70°F. Thereafter, assembly shall operate as specified herein during and after exposure at minus 25°F ambient temperature.

3.4.10.3 Humidity. Instrument panel assembly shall withstand storage, and shall operate as specified herein during and after exposure, in relative humidity of 94 percent ± 4 percent at 95° ± 2°F.

3.4.11 Waterproofness. Instrument panel assembly shall operate as specified herein, and interior shall remain dry, during and after submersion in salt water at ± 6 pounds per square inch (psi) pressure differential.

3.4.12 Shock resistance. Instrument panel assembly shall operate as specified herein during and after being subjected to shock impulses of 30 gravity units (g) intensity, of 11 ms duration, applied in each direction of each of three mutually perpendicular axes.

3.4.13 Vibration resistance. Instrument panel assembly shall withstand vibration for 90 minutes in each direction of each major axis. Vibration frequency shall be varied between 5 and 500 Hz and ambient temperature shall be maintained at 140°F.

3.5 Marking.

3.5.1 Wire marking. Each wire shall be marked with applicable circuit diagram number. Individual wires shall be marked at three-inch intervals. Wires contained in cables shall be marked within three inches of each termination. Marking shall conform to MIL-M-81594.

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3.6 Workmanship. Workmanship shall be in accordance with MIL-STD-454, requirement 9.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Materials. To determine conformance to 3.2 through 3.3.4, the contractor's inspection records shall be examined by the Government.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article (preproduction) inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions.

4.3.1 Environment. Unless otherwise specified herein, all tests shall be conducted under the following environmental conditions:

Ambient air temperature	73°F ± 18°F
Barometric pressure	28.5 (+2.0, -4.5) inches Hg
Relative humidity	50 percent ± 30 percent

4.3.2 Apparatus. Equipment listed below, or equivalent, shall be used for inspections specified herein:

a. Multimeter, Fluke 8100A
b. Power supply, Trygon T50-750, 2 required (5 amp capacity)
c. Timer, 0-150 seconds + 1 second
d. Constant current supply, HP 6177B
e. Function generator, HP 209A
f. Pulse source, 1 pulse per second nominal, pulse width 0.50 ± 0.2 seconds, amplitude 24 ± 6 Vdc

4.3.2.1 Panel circuit inspection. Panel circuits shall be inspected by means of apparatus which will provide electric power through independent panel circuits in simulation of intended use and service. Apparatus shall include resistances as required.

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4.4 First article (preproduction) inspection. Two instrument panel assemblies, produced before production begins, shall be subjected to first article (preproduction) inspection to determine conformance to requirements of this specification. Inspection shall consist of examination as specified in 4.5.2.1, and tests specified in table I.

4.4.1 First article inspection failure. Failure to pass any examination or test will be cause for refusal to grant first article approval until corrective action taken by the contractor has been approved by the Government.

TABLE I. Classification of tests.

		Test		
Characteristic	Requirement	First article	100% Acceptance	Control
Input voltage	3.4.1	4.6.1	4.6.1	
Voltage drop	3.4.2	4.6.2	4.6.2	
Current capacity	3.4.3	4.6.3	4.6.3	
Circuit resistance	3.4.4	4.6.4	4.6.4	
Circuit isolation	3.4.5	4.6.5	4.6.5	
Transient voltage characteristics	3.4.6	4.6.6	4.6.6	
Lamp circuits	3.4.7	4.6.7	4.6.7	
Independent circuits	3.4.7.1	4.6.8	4.6.8	
Dependent circuits	3.4.7.2	4.6.9	4.6.9	
LOW BATTERY CHARGE lamp	3.4.7.3	4.6.10	4.6.10	
HYDRAULIC SYSTEM MALFUNCTION lamp	3.4.7.4	4.6.11	4.6.11	
Tachometer	3.4.8.1	4.6.12	4.6.12	
Speedometer	3.4.8.2	4.6.13	4.6.13	
Fuel meter	3.4.8.3	4.6.14	4.6.14	
Voltmeter	3.4.8.4	4.6.15	4.6.15	
Fire extinguisher 2nd shot	3.4.9.1	4.6.16	4.6.16	
Fuel tank selector switch	3.4.9.2	4.6.17	4.6.17	
High temperature	3.4.10.1	4.6.18.1		4.6.18.1
Low temperature	3.4.10.2	4.6.18.2		4.6.18.2
Humidity 1/	3.4.10.3	4.6.18.3		4.6.18.3 1/
Waterproofness	3.4.11	4.6.19	4.6.19	
Shock resistance	3.4.12	4.6.20		4.6.20
Vibration resistance	3.4.13	4.6.21		4.6.21

1/ One of every three control test samples shall be humidity tested.

4.5 Quality conformance inspection.

4.5.1 Sampling.

4.5.1.1 Lot formation. A lot shall consist of all panel assemblies from an identifiable production period, from one manufacturer, to be subjected at one time to acceptance inspection.

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4.5.1.2 Sampling for examination. Samples for quality conformance examination shall be selected in accordance with MIL-STD-105.

4.5.1.3 Sampling for acceptance testing. All instrument panel assemblies (100 percent) furnished under this specification shall undergo acceptance tests.

4.5.2 Examinations.

4.5.2.1 Acceptable quality level. Instrument panel assemblies selected in accordance with 4.5.1.2 shall be examined to determine conformance to the following acceptable quality levels (AQLs) on the basis of percent defective:

Classification	AQL
Major	1.0
Minor	2.5

4.5.2.2 Classification of defects. For examination purposes, defects shall be classified as listed in table II.

TABLE II. Classification of defects.

Category	Defect	Method of inspection
Major:		
101	Dimensions affecting interchangeability not within tolerance (see 3.3).	SIE 1/
102	Improper, inadequate soldering (see 3.3.5).	Visual
103	Improperly crimped contacts (see 3.3.6).	Visual
104	Improperly crimped terminals (see 3.3.7).	Visual
Minor:		
201	Dimensions not affecting interchangeability not within tolerance (see 3.3).	SIE
202	Improper heat shrinking (see 3.3.8).	Visual
203	Improper lacing (see 3.3.9).	Visual
204	Improper marking (see 3.5.1).	Visual
205	Improper workmanship (see 3.6).	Visual

1/ SIE = Standard Inspection Equipment.

4.5.3 Classification of tests. Classification of tests shall be as follows:

a. Acceptance tests (see 4.5.4).
b. Control tests (see 4.5.5).

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4.5.4 Acceptance inspection.

4.5.4.1 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the defects specified in table II.

4.5.4.2 Tests. Each instrument panel assembly (100 percent) of each lot shall be subjected to the acceptance tests specified in table I.

4.5.4.3 Acceptance inspection failure. Failure of instrument panel assembly to pass examination or test shall be cause to reject failed unit.

4.5.4.4 Disposition of rejected units. Instrument panel assemblies, rejected as the result of tests specified in 4.5.4.2, may be repaired or reworked and resubmitted for examination and testing.

4.5.5 Control tests.

4.5.5.1 Sampling for control tests. Control test specimens shall be selected from a production lot which has passed quality conformance examination specified in 4.5.2 and acceptance tests specified in table I. Control test specimens shall be selected at the rate of one from each lot of 200 assemblies consecutively produced, except that not more than one shall be selected in each three month period, and not less than one shall be selected in each six month period. Each specimen shall be subjected to the control tests specified in table I, except that only one in three specimens shall be humidity tested.

4.5.5.2 Failure. Failure of a specimen to pass control testing will be cause for Government refusal to accept panel assemblies until corrective action taken by the contractor has been approved by the Government.

4.5.5.3 Disposition. After cause of failure has been determined and corrective action has been approved by the Government, rejected production lot shall be repaired. Three specimens shall be selected at random from repaired assemblies. One specimen shall be subjected to, and shall pass, first article inspection. Remaining two specimens shall pass inspection of those characteristics which resulted in the failure, and of those affected by the corrective action taken.

4.5.5.4 Units destroyed in testing. Panel assemblies which have been subjected to destructive tests shall be indelibly marked "DO NOT USE", and shall not be delivered to the Government.

4.6 Conformance verification.

4.6.1 Input voltage test. To determine conformance to 3.4.1, instrument panel assembly shall be correctly connected between dc power supply and ground. Supply voltage shall be connected, and then varied between 11 Vdc and 30 Vdc.

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4.6.2 Voltage drop test. To determine conformance to 3.4.2, meters shall be correctly connected to permit observation of difference between voltage from power source to instrument panel and voltage from instrument panel to ground. Circuit shall be connected to dc power source and voltages observed.

4.6.3 Current capacity test. To determine conformance to 3.4.3, instrument panel assembly shall be correctly connected between 28 Vdc power source and ground with resistive load between panel and ground to cause 250 ± 25 current draw. Simulations of operating conditions shall be applied, and results observed.

4.6.4 Circuit resistance test. To determine conformance to 3.4.4, panel assembly shall be tested in accordance with method 303, MIL-STD-202.

4.6.5 Circuit isolation test. To determine conformance to 3.4.5, panel assembly shall be tested in accordance with method 302, MIL-STD-202, using not less than 110 Vdc. Voltage shall be applied between any two isolated circuits, and between any circuit and assembly housing.

4.6.6 Transient voltage characteristics test. To determine conformance to 3.4.6, assembly shall undergo procedure specified in MIL-STD-1275.

4.6.7 Lamp circuits test. To determine conformance to 3.4.7, instrument panel assembly shall be connected to apparatus to apply correct voltage and current, of correct polarity, to appropriate panel terminals of each lamp circuit. Means shall be provided to permit observation of voltage at which lamp begins to glow, and to permit observation of changes in brightness as voltage is varied between minimum and maximum specified.

4.6.8 Independent circuits test. To determine conformance to 3.4.7.1, lamps shall be tested as specified in 4.6.7 while engine circuit is closed. Engine circuit shall then be opened, and procedure repeated.

4.6.9 Dependent circuits test. To determine conformance to 3.4.7.2, lamps shall be tested as specified in 4.6.7 while engine circuit is closed. Engine circuit shall then be opened, and procedure repeated.

4.6.10 LOW BATTERY CHARGE lamp test. To determine conformance to 3.4.7.3, panel assembly shall be correctly connected to apparatus to apply variable voltage input to lamp circuit. 24 Vdc shall be applied and lamp observed. Voltage shall be reduced to 23.1 Vdc and elapsed time to lighting of lamp shall be observed and recorded.

4.6.11 HYDRAULIC SYSTEM MALFUNCTION lamp test. To determine conformance to 3.4.7.4, panel assembly shall be correctly connected to apparatus to apply power to HYDRAULIC SYSTEM MALFUNCTION lamp circuits. Power shall be applied to function circuit and lamp observed. Power shall be simultaneously removed from function circuit and applied to malfunction circuit. Elapsed time from change of circuits to lighting of lamp shall be observed and recorded.

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4.6.12 Tachometer test. To determine conformance to 3.4.8.1, 5.0 Vdc (+0, -0.1 Vdc) shall be applied to appropriate panel terminals. Resultant dial reading shall be observed.

4.6.13 Speedometer test. To determine conformance to 3.4.8.2, sine wave input signal of 1812 Hz ± 15 Hz at 2 V peak-to-peak shall be applied to appropriate panel terminals. Resultant dial reading shall be observed.

4.6.14 Fuel meter test. To determine conformance to 3.4.8.3, 5.0 ma ± 0.1 ma current shall be applied through appropriate panel terminals. Resultant dial reading shall be observed at each position of selector switch.

4.6.15 Voltmeter test. To determine conformance to 3.4.8.4, 30 Vdc ± 0.1 Vdc shall be applied to appropriate panel terminals. Resultant dial reading shall be observed.

4.6.16 Fire extinguisher switch (second shot) test. To determine conformance to 3.4.9.1, instrument panel assembly shall be correctly connected in circuit with power supply, resistor, switch, FIRE lamp, FIRST SHOT lamp, and test voltmeters. Switch shall be actuated and voltage drop observed.

4.6.17 Fuel tank selector switch test. To determine conformance to 3.4.9.2, instrument panel assembly shall be correctly connected in circuits with power supply, variable resistors simulating tank transducers and fuel pumps, LOW FUEL LEVEL warning lamp, panel meter, and fuel tank selector switch. Switch shall be actuated and meter observed while various fuel levels are simulated in each circuit. Current applied shall be not more than 5 ma.

4.6.18 Climate test. To determine conformance to 3.4.10, instrument panel assembly shall be subjected to tests specified in 4.6.18.1, 4.6.18.2, and 4.6.18.3.

4.6.18.1 High temperature test. Instrument panel assembly shall undergo procedure I, method 501, MIL-STD-810. Pretest data is required. Highest temperature shall be 160°F. Highest operating temperature shall be 140°F.

4.6.18.2 Low temperature test. Instrument panel assembly shall undergo procedure I, method 502, MIL-STD-810. Pretest data is required. Lowest temperature shall be minus 70°F. Lowest operating temperature shall be minus 25°F.

4.6.18.3 Humidity. Instrument panel assembly shall undergo procedure II, method 507, MIL-STD-810. Pretest data is required. Measurements shall be taken during cycling.

4.6.19 Waterproofness. To determine conformance to 3.4.11, instrument panel assembly shall undergo procedure specified in method 100, procedure 1 of MIL-STD-1184. Pretest data is required.

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4.6.20 Shock resistance test. To determine conformance to 3.4.12, instrument panel assembly shall be subjected to procedure I, method 516, MIL-STD-810. Pretest data is required. Configuration of pulse shall be half sine wave of 30 g ± 3 g peak amplitude and 11 ms duration. There shall be three pulses in each direction on each axis (total 18 pulses). Panel assembly shall be operated during test. Aftertest data is required.

4.6.21 Vibration resistance test. To determine conformance to 3.4.13, instrument panel assembly shall be mounted on vibration test apparatus as in intended use, and electrically connected to make all functions operable. Ambient temperature shall be maintained at 140°F during vibration testing. Panel assembly shall be subjected to vibration as specified in procedure VIII, method 514, MIL-STD-810, except cycling time per axis shall be 180 minutes, and vibration levels shall conform to table III.

TABLE III. Vibration levels.

Axis	Frequency (Hz)	Amplitude
Vertical	5 to 25	± 1.0 g
	25 to 57	0.030 inch D.A.
	57 to 500	± 5.0 g
Longitudinal and Latitudinal	5 to 25	± 1.0 g
	25 to 44	0.030 inch D.A.
	44 to 500	± 3.0 g

Test procedures 4.6.1 through 4.6.17 shall be repeated during, and after, vibration testing in 140°F ambient air temperature. Subsequently, air temperature shall be reduced to room ambient, and procedures 4.6.1 through 4.6.17 shall be repeated.

5. PACKAGING

5.1 Preservation, packaging, packing, and marking. Preservation, packaging, packing, and marking for the desired level of protection shall be in accordance with the applicable packaging standard or packaging data sheet specified by the contracting aurhority (see 6.2).

6. NOTES

6.1 Intended use. Instrument panels furnished under this specification are intended for use in driver's compartment of M1 tank in military service under specified conditions of climate and terrain.

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6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Applicable level of packaging, packing, and marking (see 5.1).

6.3 First article. First article samples shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for examination, tests, and approval of the first article (see 3.1).

6.4 Recycled materials. The use of recycled materials which meet the requirements of the applicable material specifications without jeopardizing the intended use of the item shall be encouraged (see 3.2).

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